Anchored Shell-Zonotope Reachability Obstruction

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Abstract

The anchored shell-prefix obstruction of RH-244 could have been caused by the prefix order. We remove that possibility in the frozen candidate window. Every conjugate-complete shell may be selected independently, and we solve the full convex box relaxation. An explicit LP dual certifies that the RH-243 deterministic anchor is unreachable at all 32 archived endpoints. The result is a finite zonotope obstruction, not a statement about expanded windows or signed spectral groupings.

1 Shell power zonotope

Let $b_n = \operatorname{tr} A^n - p^n - q^n$ and let a_n be the RH-243 Hardy-scaled anchor [3]. For each conjugate-complete shell S_j in the RH-222 candidate window define the real power vector

$$v_j = \Big(\sum_{s \in S_j} s^n\Big)_{n=2}^{12}, \qquad d = (b_n - a_n)_{n=2}^{12}.$$

Conjugacy makes these vectors real; the archived imaginary leakage is zero to floating precision. A shell subset uses $w_j \in \{0,1\}$, while its box relaxation allows $0 \le w_j \le 1$. With weights $\omega_n = 1/n$, set

$$\delta_\square = \min_{0 \le w \le 1} \sum_{n=2}^{12} \omega_n \left| d_n - \sum_j w_j v_{j,n} \right|. \tag{1}$$

Theorem 1 (zonotope dual certificate)**.** *For the finite real matrix $V = (v_{j,n})$,*

$$\delta_\square = \max_{|y_n| \le \omega_n} \left[y^T d - \sum_j \max(0, y^T v_j) \right]. \tag{2}$$

Consequently, any feasible y whose displayed value exceeds a tolerance already excludes every shell prefix, every contiguous shell interval, and every binary shell subset.

Proof. Use $|x| = \max_{|y| \le 1} yx$ coordinatewise, absorb the weights into the constraints $|y_n| \le \omega_n$, and interchange the finite minimization and maximization. For fixed y, minimizing $-y^T V w$ over the box gives $-\sum_j \max(0, y^T v_j)$. Strong finite-dimensional LP duality proves (2). □

2 Audit

We reconstruct all 32 RH-222/RH-236 endpoints and solve both primal and dual LPs, in orders 2–12 and with tolerance $\varepsilon_\sigma = \sigma$ [1, 2]. The aggregate candidate counts are

543 prefixes, 5012 contiguous intervals, 139572890 binary subsets of rank at least four.

The exact dual lower bounds give zero passes for all three discrete classes and for the entire box relaxation.

selection class	candidates	passes	best-distance range
prefix	543	0/32	0.397238–0.484576
contiguous interval	5012	0/32	0.397238–0.484576
binary shell subset	139572890	0/32	0.168577–0.424018
box relaxation	–	0/32	0.146498–0.424018

The minimum box failure margin is 0.14524763462315904, the minimum distance/tolerance ratio is 10.17255613514909, and the maximum is 118.76033343311221. The largest primal–dual gap is 7.72×10^{-15}. Every best interval begins at the outer shell. A single-coordinate certificate suffices at 24 endpoints (18 at order 2 and 6 at order 6); the other eight require the multi-order dual. The effective box ranks range from 12 to 31.430747812062542.

3 Boundary

The theorem and audit concern only the frozen resolved candidate windows, complete conjugate shells used at most once, orders 2–12, real conjugate power vectors, and the RH-238 tolerance. They do not exclude deeper or additional roots, expanded windows, signed/complex grouping, continuum mechanisms, or unbounded shell multiplicities. In particular, a fractional box weight is a reachability certificate, not a legal spectral multiplicity.

The deterministic anchor is therefore not available inside the current single-use shell class. The next test may relax the upper bound on weights, while monitoring whether the required multiplicities cease to represent a finite algebraic cloud. Gate A remains open and Gates B–E are untouched. No Hilbert–Polya operator, zeta-divisor equality, Riemann-zero identification, or RH implication is claimed.

References

[1] Bin Wang. Rank-growing conjugate cloud atlas. 2026. RH-222 preprint.

[2] Bin Wang. Cloud-extracted trace moment atlas. 2026. RH-236 preprint.

[3] Bin Wang. Deterministic numerator coefficient anchor dictionary. 2026. RH-243 preprint.